Course Syllabus

1. Course Number: 2110531
2. Course Credit: 3 (3-0-6)
3. Course Title: Data Science and Data Engineering Tools
4. Department Computer Engineering
5. Semester First Semester
6. Academic Year 2025
7. Instructor Peerapon Vateekul, Ph.D.
   Natawut Nupairoj, Ph.D.
   Veera Muangsin, Ph.D.
8. Condition -
9. Status Elective
10. Curriculum Computer Engineering
11. Degree M.Eng., M.Sc.
12. Hours/Week 3-Hour Lecture & Lab

13. Course Description

Data Science (DS) is the study of the discovery of knowledge from data. Being a data scientist requires an integrated skill set spanning mathematics, statistics, machine learning, databases, and other branches of computer science along with a good understanding of the craft of problem. Data Engineering (DE) is the study of how to engineer or process data, i.e., data cleansing, data storing, etc. There are three main parts in this course:

- Data engineering: Data exploration & preparation
- Data science: Machine Learning techniques
- Data visualization: Storytelling via data

14. Course Outline

14.1. Learning Objectives

- Describe what DS and DE are and the skill sets needed
- To be able to explore and understand collected data
- To be able to analyze data by apply traditional machine learning techniques
- To be able to visualize data in relation to spatial and temporal points of views

14.2. Learning Contents

- Section1 on Tue 1PM-4PM, Building ENG3 Room 417
- Section5 on Sat 1PM-4PM, Building ENG4 Room 18-16
- *** Please bring your laptop to the class ***
- Students need to attend the class on-site at least 80% (at least 12 weeks) as a mandatory criterion to "pass" this course.

#	Tue 1PM-4PM	Sat 1PM-4PM	Contents	Instructor	Module
1	05-Aug-25	09-Aug-25	Introduction, Pandas	Aj.Peerapon	DS1
2	12-Aug-25	16-Aug-25	Data Preparation for ML	Aj.Peerapon	DS2
3	19-Aug-25	23-Aug-25	Traditional ML (1)	Aj.Peerapon	DS3
4	26-Aug-25	30-Aug-25	Traditional ML (2)	Aj.Peerapon	DS4
5	02-Sep-25	06-Sep-25	Deep Learning (1); CNN, RNN (LSTM, GRU)	Aj.Peerapon	DS5
6	09-Sep-25	13-Sep-25	Deep Learning (2); Transformer	Aj.Peerapon	DS6
7	16-Sep-25	20-Sep-25	Text Classification & LLM	Aj.Peerapon	DS7
7	23-Sep-25	27-Sep-25	Midterm Exam Week (22 - 26 Sep 2025)		
8	30-Sep-25 Online	04-Oct-25	Data visualization (Graduation Week: 29,30 Sep & 1 Oct 2025)	Aj.Veera	VIZ1
9	07-Oct-25	11-Oct-25	Python visualization	Aj.Veera	VIZ2
10	14-Oct-25	18-Oct-25	Graph analysis & spatial analysis	Aj.Veera	VIZ3
11	21-Oct-25	25-Oct-25	Big data architecture + data storage	Aj.Natawut	DE1
12	28-Oct-25	01-Nov-25	Web scraping	Aj.Natawut	DE2
13	04-Nov-25	08-Nov-25	Data ingestion	Aj.Natawut	DE3
14	11-Nov-25	15-Nov-25	Big data processing (Spark)	Aj.Natawut	DE4
15	18-Nov-25	22-Nov-25	MLOps: Orchestration (Airflow) and serving (FastAPI, Seldon Core)	Aj.Natawut	DE5
	25-Nov-25	29-Nov-25	Final Exam Week (24 Nov - 8 Dec 2025) *** Sat 29 Nov 2025 at 1PM-4PM ***		

* There will be up to two guest speakers in the class.

14.3. Method: Lecture and Lab
14.4. Learning Media: PowerPoint presentation, Zoom
14.5. Evaluation

- Module1 Assignment (data science) 10%
- Module2 Assignment (data engineering) 10%
- Module3 Assignment (data visualization) 5%
- Midterm Exam (Kaggle) 20%
- Project 20%
- Attendance 5%
- Final Exam 30% (Lab Test)

15. Reading List
    15.1. Required Text: N/A
    15.2. Electronic Media or Websites:

16. LMS
    16.1. CourseVille: "Agentic"
    16.2. Discord: https://discord.gg/v7AqAHZAJt
    16.3. GitHub: https://github.com/pvateekul/2110531_DSDE_2025s1